LESSON 10.2

Math is Beauty

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use bilateral symmetry to describe structure in nature and in human creations.

WHY THIS MATTERS There are over 17,000 species of butterflies in the world, and most people are fascinated by them. Part of the fascination is mathematical: math creates beautiful designs, many of which are found naturally in plants and animals.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
symmetric *Spanish: simétrico*	Having two halves that match — one side is a mirror image of the other.	A butterfly: the left wing is the mirror of the right.
mirror image *Spanish: imagen reflejada*	A copy of a figure flipped across a line, so the two halves face each other.	Fold the layer down the middle: each leaf lands on its partner.
bilateral symmetry *Spanish: simetría bilateral*	Reflective symmetry in plants and animals — the two sides of the body mirror each other.	All 17,000+ butterfly species have it — two sides, one mirror line.
line of symmetry *Spanish: eje de simetría*	The line you can draw through a figure so each side is a reflection of the other.	The line straight down the butterfly's body.
balance *Spanish: equilibrio*	An even, steady arrangement — neither side feels heavier than the other.	Two matching halves — neither side feels heavier than the other.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I look at the two sides of the butterfly, and they look like mirror images — the butterfly is symmetric.

1. All butterflies have reflective symmetry. In plants and animals, this is called bilateral symmetry.
2. To show it, I draw a line down the middle of the butterfly's body: everything on the left is reflected on the right.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

This plant grows in layers, and each layer is made up of two leaves — so 3 layers hold 3 × 2 = 6 leaves.

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ has a line of symmetry _____ because the two halves are _____.
- I have seen symmetry in _____ .
- The two sides _____ .
- A _____ has bilateral symmetry.

Word bank symmetric | mirror image | bilateral symmetry | line of symmetry | balance

WATCH OUT!

A common mistake

Thinking a figure lacks bilateral symmetry just because its line of symmetry is not vertical

LESSON 10.2

Math is Beauty

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I look at the two sides of the butterfly, and they look like mirror images — the butterfly is symmetric.

1. All butterflies have reflective symmetry. In plants and animals, this is called bilateral symmetry.
2. To show it, I draw a line down the middle of the butterfly's body: everything on the left is reflected on the right.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

What does it mean that a butterfly is symmetric?

(A) The butterfly has exactly two wings

(B) The two sides of the butterfly look like mirror images

(C) The butterfly's wings are very colorful

(D) The butterfly is larger than most insects

STEPS TO TRY
1. Think about what you see on the left wing versus the right wing.
2. Imagine a mirror standing along the butterfly's body.
3. What word describes two sides that reflect each other?

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

In plants and animals, reflective symmetry is called:

(A) rotational symmetry

(B) a mirror image

(C) bilateral symmetry

(D) balance

STEPS TO TRY

1. The word starts with 'bi-', meaning two.
2. It describes the two sides of a body.
3. Bi-lateral means two-sided.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A plant grows in layers, and each layer is made up of 2 leaves. How many leaves does the plant have after growing 4 layers?

(A) 4 leaves

(B) 6 leaves

(C) 8 leaves

(D) 16 leaves

STEPS TO TRY

1. How many leaves are in ONE layer?
2. There are 4 equal layers.
3. Multiply 4 × 2.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

Where have you seen symmetry before? Name one place and describe it.

SENTENCE STARTER I have seen symmetry in ___ .
The two sides ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find something in this room that you think has bilateral symmetry and your partner thinks does not. Where exactly would the line of symmetry go, and what makes the two halves mirror images rather than just similar?

SENTENCE STARTER ___ has a line of symmetry ___ because the two halves are ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help